

HOW ADAPTIVE ENGINE WORKS:

BKT:

What BKT and the DAG Should Do

- **Bayesian Knowledge Tracing (BKT):** Maintains a continuous probabilistic belief state $P(L_t) \in [0.0, 1.0]$ representing the likelihood that a learner has mastered a specific concept. It updates after every question attempt by mathematically separating genuine understanding from lucky guesses or careless slips.
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- **Directed Acyclic Graph (DAG):** Represents the curriculum hierarchy as ordered nodes with prerequisite dependencies. It acts as a structural gatekeeper, preventing learners from progressing to complex downstream concepts until upstream foundational nodes are mastered.
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How Initial Weights and Parameters Are Determined

- **$P(L_0)$ (Initial Prior = 0.10):** Set as a standard cold-start baseline in Intelligent Tutoring Systems (ITS) literature, assuming a learner has low initial knowledge of a new concept prior to diagnostic evidence.
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- **$P(T)$ (Transition Probability = 0.20):** Represents the estimated probability that a learner acquires understanding of the concept during a practice/instruction step.
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- **$P(G)$ (Guess Probability = 0.20):** Accounts for multiple-choice noise, reflecting the baseline odds of guessing the correct option without true mastery.
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- **$P(S)$ (Slip Probability = 0.10):** Represents the probability that a knowledgeable student makes an accidental mistake or typo.
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- **Phase 2 Parameter Calibration:** After baseline testing, parameters are refined offline across the cohort logs using Expectation-Maximization (EM) or grid search to fit the exact empirical difficulty of each item.
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How Learning Happens in the Engine

1. **Diagnostic Initialization:** The student takes a baseline test; correct items initialize that concept node to $P(L_0) = 0.85$, while incorrect items remain at $P(L_0) = 0.10$.
- 2.
3. **Posterior Belief Update:** When an answer is submitted, Bayes' Rule updates the state based on whether the result was correct or incorrect:
- 4.

- *If Correct:* $P(L_t \mid \text{Correct}) = \frac{P(L_t)(1 - P(S))}{P(L_t)(1 - P(S)) + (1 - P(L_t))P(G)}$
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 - *If Incorrect:* $P(L_t \mid \text{Wrong}) = \frac{P(L_t)P(S)}{P(L_t)P(S) + (1 - P(L_t))(1 - P(G))}$
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5. **Knowledge Transition Step:** The engine predicts the learner's updated readiness for the next step by factoring in learning gain: $P(L_{t+1}) = P(L_t \mid \text{Obs}) + (1 - P(L_t \mid \text{Obs})) \cdot P(T)$. If a hint was revealed, $P(T)$ is reduced to account for scaffolding assistance.

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Use Cases and Dynamic Navigation

● Use Case 1: Fast Mastery Progression

- *Condition:* Learner answers standard questions correctly without hints.
- *Navigation:* Mastery climbs ($0.10 \rightarrow 0.59 \rightarrow 0.89$). Upon crossing $P(L) \geq 0.85$, the engine marks the node mastered and traverses the DAG forward to the next prerequisite-cleared concept.

● Use Case 2: In-Node Scaffolding (Struggling Learner)

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- *Condition:* Learner misses a question or exhibits high response latency, causing $P(L) < 0.40$.
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- *Navigation:* Engine keeps the learner on the same DAG node, swaps dense instructional text for a simplified explanation, and delivers an easier recall question.

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● Use Case 3: Upstream Prerequisite Remediation

- *Condition:* Learner persistently fails questions on a downstream concept due to an unmastered parent node.
- *Navigation:* The engine traverses backward along the DAG edges to the parent prerequisite concept, serving foundational review content before returning to the original topic.

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● Use Case 4: Safety Escalation

- *Condition:* Learner encounters 5 consecutive failures despite hints and simplified explanations.
- *Navigation:* Engine halts repeated drill loops, displays a supportive pause message, and logs an escalation event in the database for admin review.

We bring in novelty of using hybrid layer:

If the student is looping in some state between pregenerated states example answers - easy question but still struggles standard question then hybrid activates - understand why they fail and guide accordingly.

There is a fallback mechanism as well

No, you never repeat the exact same easy explanation and question.

Repeating identical content causes student memorization rather than learning. Instead, the adaptive engine steps through a **graduated pedagogical sequence** across distinct pre-generated artifacts:

SCAFFOLDING STEP-SEQUENCE (Within a Node)	
1st Error	→ Reveal `hint` on standard question
2nd Error	→ Serve `simple_explanation` + `easy_question`
3rd Error	→ Serve `example` (interest-based) + 2nd easy question variant
4th Error	→ Serve `prerequisite` micro-lesson + bridge question[cite: 2]
5th Error	→ Lock exercise and trigger `escalations` to teacher[cite: 2]

The engine adapts to an individual learner by maintaining a **dynamic, multi-dimensional student profile** that continuously updates after every single click, response, and hesitation.

Adaptation operates across **4 distinct adaptive vectors**:

1. Mathematical Mastery Tracking (BKT Vector)

The engine does not grade students with static percentages. It maintains an active posterior probability of mastery ($P(L)$) for every concept node in the curriculum graph:

- **Distinguishing Slips from Guesses:** If a student who previously demonstrated high mastery misses one question, the Bayesian update treats it as a likely slip ($P(S) = 0.10$) rather than dropping their progress to zero.
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- **Independent Mastery Acceleration:** When a learner answers challenging questions correctly without scaffolding, their $P(L)$ climbs directly toward ≥ 0.85 , allowing them to skip repetitive drilling and advance quickly.

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2. Cognitive Load & Error-State Adaptation (Scaffolding Vector)

The engine tracks real-time interaction patterns (consecutive errors, hint requests, response latency) to adapt the **modality and density of the content**:

- **1st Hesitation / Wrong Answer:** Keeps standard question difficulty, but unlocks an accordion-style reasoning hint.
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- **2nd Wrong Answer:** Detects high cognitive friction; automatically replaces dense theoretical text with a **simple_explanation** (shorter sentences, plain vocabulary) and drops question difficulty to direct recall (**easy_question**).
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- **3rd Wrong Answer:** Injects real-world analogies (**example**) tied to the student's selected interest.
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- **Persistent Struggle (≥ 5 Errors):** Halts the drill loop entirely and flags an escalation log for human instructor review to prevent frustration.

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3. Structural Curriculum Routing (Prerequisite DAG Vector)

Instead of following a linear, rigid syllabus, the student navigates a Directed Acyclic Graph (DAG):

- **Pre-Diagnostic Skipping:** If a student answers a concept correctly in the 4-question onboarding diagnostic, that node is marked mastered ($P(L_0) = 0.85$), allowing them to jump straight to their actual knowledge frontier.
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- **Backward Prerequisite Remediation:** If a student struggles on an advanced topic (e.g., *Loops*) because they lack foundational knowledge (e.g., *Conditions*), the engine detects the gap and branches backward to review the upstream parent concept before returning to the original lesson.

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4. Thematic & Interest Contextualization (Presentation Vector)

During onboarding, the learner selects an interest domain (such as *Gaming*, *Sports*, *Music*, or *Cartoon Animals*):

- When abstract concepts are introduced or when remediation is triggered, the engine adapts instructional examples to match that theme.
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- **Example (Variables & Iteration):** A gaming-oriented profile receives examples referencing player health bars, inventory slots, and level counters, while a

sports-oriented profile receives examples referencing scoreboard updates and lap timers.

Summary: Profile-Driven State Machine

Student Profile Signal	Engine Observation	Real-Time Adaptation Triggered
High Diagnostic Score	Fast baseline comprehension	Skips introductory lessons; targets frontier concepts.
Consecutive Correct Answers	$P(L) \geq 0.70$	Serves hard_question to confirm deep synthesis.
Repeated Misconceptions	$P(L) < 0.40$	Activates simplified micro-lessons and hints.
Missing Prerequisite	Parent node $P(L) < 0.85$	Temporarily redirects to foundational review.
Declared Interest	Tag = <i>Gaming / Sports</i>	Contextualizes analogies to that thematic domain.